

Synchronising Sensors With Geotagged Video

Rob Smith
26 March 2025



Away Team Software

www.awayteam.co.uk



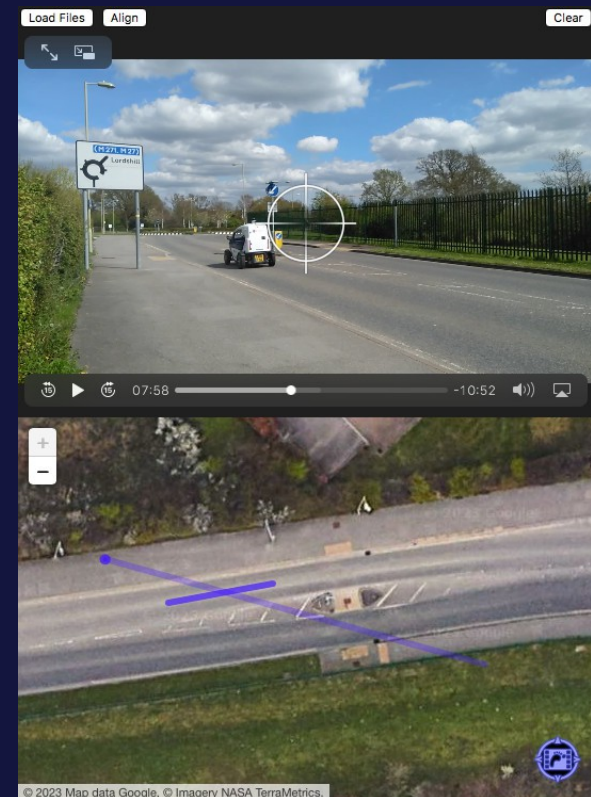
WebVMT Overview

- **Enabling Technology**
 - Open web format for location sync with video
- **Designed For The Web**
 - Sharing, presenting & indexing geospatial data
- **Engagement Website: webvmt.org**
 - Overview, technical demos & community tools
- **W3C Group Note (September 2023)**
 - Use cases, syntax examples & draft specification
 - <https://www.w3.org/TR/webvmt/>



OGC Testbed 19

- **Wrong-way Traffic Use Case**
 - Synchronise with sight lines
 - Create detection zones
 - Interpolate path
 - https://youtu.be/RrF2NUCg_cM
- **Device Synchronisation**
 - Two smartphone cameras
 - Accurate to within 100ms





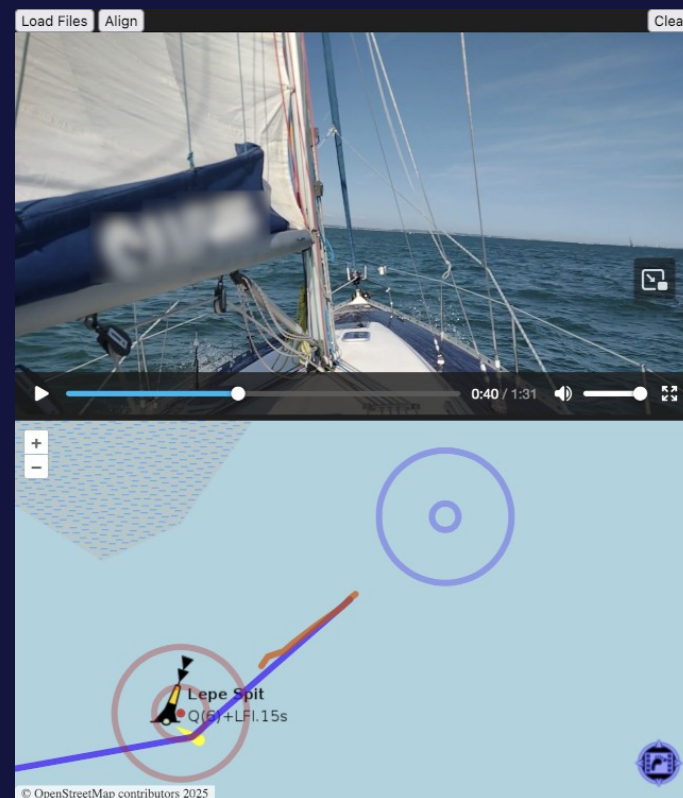
OGC Testbed 20

- **Maritime Use Case**

- On-board geotagged video
- Sync with AIS tracking data
- Aggregate sensor data

- **Aerial Observation Use Case**

- Export in-band metadata
- Calculate camera heading
- Control map orientation





WebVMT Prototypes

- **Negative Cue Times**
 - For non-destructive synchronisation
- **Simplified Data Interpolation**
 - Equivalent to path interpolation
- **Heading**
 - Location can optionally have direction
 - Includes interpolation
- **Map Initialisation Event**
 - Custom HTML event for map start-up



Proposed Enhancements

- **WICG DataCue: github.com/WICG/datacue**
 - Access timed metadata in HTML/Javascript
 - Synchronise & search sensor data
- **Web Map APIs**
 - Identify new requirements & use cases
- **3D Features**
 - 2D extrusions & 3D models
- **IANA Type: `text/vmt`**
 - Breaking changes to `text/vtt`



Breakout Discussion

- **Identify New Use Cases**
 - For dash cams, drones, body cams & phones
 - With sensor synchronisation & aggregation
- **Suggest Enhancements**
 - Compelling use cases?
- **Discuss Web Interfaces & Interoperability**
 - Requirements for web map APIs
 - DataCue & custom TextTrackCue
- **Summarise Conclusions**



Next Steps

- **Raise Issues In WebVMT Community Group**
 - Geotagged video use cases with sensors
 - Web map APIs & DataCue API
 - <https://github.com/webvmt/community-group/>
- **Make Your Own Geotagged Videos**
 - Get Started guidance
 - <https://webvmt.org/community/get-started>
- **Share Your Experiences**